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Spotlight on Artificial Intelligence and Freedom of Expression

A Policy Manual

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AI in Content Curation

AI in Content Curation and Media Pluralism

This part focuses on the use of AI in content curation, addressing the impact of data-driven content recommender systems on diversity and media pluralism. This part and the next one highlighing shortcomings of AI-based content curation and targeted advertising provide human rights-centred recommendations to prevent the negative impact of AI tools in content curation on the right to freedom of opinion and expression.

1. Defining the scope of content curation's impact on media pluralism

1.1 The relevance of algorithmic content curation and data-driven recommendation systems to media pluralism and diversity

Diversity and media pluralism are core democratic principles whose quality is impacted by the rise of dominant internet intermediaries and their influence over public discourse. Internet intermediaries, in particular social media platforms, have become an important source of, access point for and key distributor of information, including news content. Information dissemination and, increasingly, aggregation occurs primarily through algorithmic content curation[26] and recommender systems. Using optimisation and analysis of human and non-human agents, these systems "deliver" personalised content customised to individual profiles, resulting in the type and amount of content to which each individual is exposed. Content recommender systems, which rank content to determine what is presented to individual users, impact individuals' freedom to seek and impart information, as well as the overall information landscape and media freedom. The design of recommender systems significantly affects what is seen online, and what remains hidden—and for whom. The

26 Content curation can be understood as a set of algorithmic and human-driven processes that support the distribution of content to audiences, such as content ranking or editorial data analysis. See at: B. Bukovska et al, Spotlight on Artificial Intelligence and Freedom of Expression #SAIFE (2020), p.19.

process of algorithmic curation is underpinned [27] by the values and goals of the algorithm's creator,[28] socio-technical factors, self-regulation (Terms of Service, for example) and state regulation. Given how ubiquitous online content has become, and the significance it has in shaping opinion and decision-making, the pivotal question arises: Where does responsibility lie in defining and implementing policy to prioritise and codify media pluralism and diversity[29] in the era of digital information?

The part of the report provides a conceptual summary of key algorithmic-curation processes and their transformative impact on media pluralism. It further provides a set of recommendations for OSCE participating States on a human-rights-centred approach to algorithmic content curation. As such, this part focuses on the impact that algorithmic content curation and data-driven recommendation systems have on media pluralism and diversity in democratic societies—and the state's role to act as the ultimate guarantor of the human right to freedom of expression and to ensure an enabling environment for this expression.

1.2 Incongruence of algorithmic content curation and freedom of expression

The ability to filter, prioritise and engage with online content based on personal preferences and interests is often at odds with the individual agency to seek, receive and impart diverse information.[30] As a basic principle, internet intermediaries typically prioritise and display content to an individual based on the system's prediction that the individual is

27 K. Klonick, The New Governors: The People, Rules, and Processes Governing Online Speech, The Harvard Law Review, p.1664.
28 Radsch, Courtney. "Digital Information Access." In A New Global Agenda: Priorities, Practices, and Pathways of the International Community, edited by D. Ayton-Shenker, 72–83. Rowman & Littlefield Publishers, 2018. https://books.google.com/books?id=tyJJ-DwAAQBAJ.
29 On exposure see: Philip M Napoli, "Rethinking Program Diversity Assessment: An Audience-Centered Approach" (1997) 10 Journal of Media Economics 59-74.; N. Helberger & M. Wojcieszak (2018). Exposure Diversity. In P. M. Napoli (Ed.), Mediated Communication (pp. 535-560). (Handbooks of Communication Science; Vol. 7). De Gruyter Mouton. https://doi.org/10.1515/9783110481129-029.
30 P. Leersen, The Soap Box as a Black Box: Regulating Transparency in Social Media Recommender Systems, European Journal of Law and Technology (2020), p.12.

likely to engage with that content. Similar to systems for personalised and behaviour-based advertisements, content recommender systems thus extensively collect data of users (and non-users) to create digital profiles, assess similarities and make inferences based on this data.

Many online platforms' business model,[31] which prioritises engagement and profit over a human rights-centred approach, can and does result in exploitative and intrusive data practices, the spread of mis/disinformation and algorithmic feedback loops.[32] It has been proven to have a negative influence on content plurality, especially regarding content created by or for marginalised communities. The model perpetuates information gaps[33] and constitutes obstacles to advocacy, thereby recreating and bolstering structural societal inequality. There is also evidence that suggests that the process of content moderation benefits those groups already dominating online spaces and narratives over marginalised groups, information and narratives.[34] Moreover, algorithm-driven content discovery (e.g. search engines) have been found to reinforce racism by suggesting discriminatory search phrases and discrepancies, particularly along racial, language and gendered lines, in depictions of members of marginalised communities.[35]

For the most part, algorithmic content curation and recommender systems are based on intermediaries' own (internal) rules, interests and assumptions, rather than democratic or public interest values.[36]

31 See: Ranking Digital Rights, It's the Business Model: How Big Tech's Profit Machine is Distorting the Public Sphere and Threatening Democracy (2020).
32 Bodó, B., Helberger, N., Eskens, S., & Möller, J, Interested in diversity: The role of user attitudes, algorithmic feedback loops, and policy in news personalization. Digital Journalism (2019), p.219.
33 A. Causevic and A. Sengupta, Whose Knowledge Is Online? Practices of Epistemic Justice for a Digital New Deal, IT for Change (2020), Retrieved from: https://itforchange.net/digital-new-deal/2020/10/30/whose-knowledge-is-online-practices-of-epistemic-justice-for-a-digital-new-deal/.
34 B. Marshall, Algorithmic misogynoir in content moderation practice, Heinrich-Böll-Stiftung (2021), p.7,11. See also: M. E, Mazzoli and D. Tambini, Prioritisation uncovered: The Discoverability of Public Interest Content Online. Council of Europe (2020), p. 44.
35 Safiya Umoja Noble, Algorithms of Oppression How Search Engines Reinforce Racism, NYU Press (2018).
36 C. Radsch. "Digital Information Access." In A New Global Agenda: Priorities, Practices, and Pathways of the International Community, edited by D. Ayton-Shenker, 72–83. Rowman & Littlefield Publishers, 2018. https://books.google.com/books?id=tyJJDwAAQBAJ.

Content recommendation is crucial for the growth and dominance of large internet intermediaries, and it lies at the heart of their business models. As recommender systems are "a key logic governing the flows of information on which we depend",[37] internet intermediaries are enabled to act as gatekeepers of information and knowledge. This has broader implications for public interest, representation and power (in)equality, on- and offline.[38] Intermediaries' recommender systems have significantly reconfigured the logic of public communication, including access to news, critical information and overall content in the public interest. Thus, their recommender systems significantly restrict equal access to, and of, journalists and media outlets, while pressuring professional journalism due to the outflow of advertising money to intermediaries. Recent research findings on algorithmic prioritisation, defined as "the range of design and algorithmic decisions that result in prominence and discoverability of content" [39]reveal the potential for the polarisation of opinions and attitudes online. For instance, an important factor in content prioritisation processes is individual political predisposition and/or affiliation. Prioritisation can, therefore, reinforce and perpetuate polarisation of opinions and attitudes online, especially among those users at the edges of the political spectrum who likely already consume a predominance of affiliated content.[40] It has also been shown that "some groups in society are more prone to selective exposure than others".[41]

37 T. Gillespie (2018). Custodians of the internet. Retrieved from https://www.researchgate.net/publication/327186182_Custodians_of_the_internet_Platforms_content_moderation_a nd_the_hidden_decisions_that_shape_social_media

38 P. Leerssen, The Soap Box as a Black Box: Regulating Transparency in Social Media Recommender Systems. Retrieved from file:///Users/eliskapirkova/Downloads/Leerssen%20EJLT_corr.pdf.

39 M.E. Mazzoli and D. Tambini. Prioritisation uncovered: The Discoverability of Public Interest Content Online. Council of Europe (2020), p.12.

40 B. Stark, D. Stegmann, Are Algorithms a Threat to Democracy? The Rise of Intermediaries: A Challenge for Public Discourse. Retrieved from https://algorithmwatch.org/wp-content/uploads/2020/05/Governing-Platforms-communications-study-Stark-May-2020 -AlgorithmWatch.pdf

41 B. Bodó, N. Helberger, S. Eskens & J. Möller, Interested in diversity: The role of user attitudes, algorithmic feedback loops, and policy in news personalization. Digital Journalism (2019), p.15.

While active personalisation based on user input tends to produce a greater diversity of information, passive personalisation based on algorithmic content selection tends to exacerbate the so-called filter bubble effect.[42]

Bias and discrimination, including gender-based discrimination, in data-supported algorithmic decision-making can occur for several reasons and at many levels in content curation systems, and they can be difficult to detect and mitigate. It has been suggested that the exclusion of sensitive/ identity-based information sufficiently protects against discrimination. Yet discrimination can and does occur, despite these "protections", given the expansive and diverse information contained in algorithm-informing datasets. Bias in algorithms can stem from design and implementation, including unrepresentative or incomplete training data, or reliance on individual, experiential, or values-informed data that reflects historical/ structural inequalities. Algorithmic bias can have a collective, disparate impact on communities, especially marginalised groups, even when there is no intention to discriminate. An exploration of both intended and unintended consequences of algorithms is thus necessary. Current public policies may not be sufficient to identify, mitigate, and remedy the impact on individuals or society at large. In addition to deliberate efforts to shape individual attention (direct manipulation), there is also a danger of unwanted and indirect biases being introduced into the algorithm through incorporation of big data at various levels of the content curation. Both direct and indirect discrimination caused by algorithms using big data are among the most pressing dangers of algorithm-driven curation processes.

The combined effect of content filtering and personalisation practically creates layers of restriction in terms of discoverability, and thus accessibility, of diverse media content. The aforementioned issues have serious implications for media pluralism, understood as a plurality of information sources (external pluralism), and of content (internal pluralism).[43] More specifically, and in the context of this report, media

42 D. Wagner, Artificial Intelligence and Disinformation as a Multilateral Policy Challenge https://www.osce.org/files/f/documents/d/0/506702.pdf.
43 On exposure see: P. M. Napoli, "Rethinking Program Diversity Assessment: An Audience-Centered Approach" (1997) 10 Journal of Media Economics 59-74.; Helberger, N., & Wojcieszak, M. (2018). Exposure Diversity. In P. M. Napoli (Ed.), Mediated Communication (pp. 535-560). (Handbooks of Communication Science; Vol. 7). De Gruyter Mouton. https://doi.org/10.1515/9783110481129-029.

pluralism also refers to the distribution of communicative power (or "voice") in society. A fair distribution of "voice", as a precondition, requires the deconcentration of power and decentralisation of resources within the information ecosystem,[44] as well as support for alternative models that offer a diversity of narratives and content. It is clear that algorithmically driven content curation processes are transforming the notions of media pluralism and diversity, which are necessary for democratic, public debate and inclusive societies.

It is against this background that state and non-state actors, primarily internet intermediaries and media organisations, but also international and regional organisations, civil society representatives and academia, are called upon to adopt policies that contribute to an enabling environment for media plurality. This means enabling access, availability, discoverability and consumption of different kinds of (media) content through different mediums and via multiple channels.

2. Algorithmic content curation and data-driven recommendation systems: impact on media pluralism

2.1 Typology

A distinct source of influence, and thus communicative power, of internet intermediaries and social media companies lies in their content recommender systems, which also "lend gravitas to their role in democratic culture".[45] In essence, a "recommender system" includes various technologies that filter, retrieve and organise information for individuals. The factors for ranking can include the level of engagement with the specific content, the type of content, when it was first shared, or how users have interacted previously with similar content. By ranking content, these systems have the potential to shape and impact individuals' ability to form opinions.

44 M. Moore and D. Tambini (eds) (2018) Digital Dominance: The Power of Google, Amazon, Facebook, and Apple. New York: Oxford University Press.
45 K. Klonick, The New Governors: The People, Rules, and Processes Governing Online Speech, The Harvard Law Review, p.1663.

The main purpose of content recommenders is to filter large amounts of information online. This algorithmically driven process works in different ways:

- **Content-based filtering**: individuals get content recommendations based on their stated or implied preferences. For example, if someone likes classical music or news about a favourite sports team, then the recommender system will prioritise these items that align with their interests, and will, likely, encourage engagement.

- **Collaborative filtering**: individuals get content recommendations based on people with whom they are closely associated or with whom they share similarities (in demographic category, content preferences, etc.). For example, when reading news, a system recommends articles a friend has shared/read, or, when doing online shopping, the system recommends items that people with a similar shopping history have purchased.

- **Hybrid filtering**: a combination of the above-mentioned filtering and curation methods. For example, recommending a news article a friend has liked, but only if it covers a certain topic of perceived interest to the user, and combining it with a wide range of different metadata, such as an individual's location, usage history, etc.

All of these processes are based on users' data, profiles and interactions with a given platform, as well as information gleaned from the underlying ad tech architecture. The algorithm specifies the precise way in which content recommendations are generated, using content-based and collaborative filtering. The system creates a recommender strategy for how data is combined to calculate potential engagement, based on user recommendations, to satisfy optimisation criteria. Put simply, algorithmic content curation is the strategy used by a recommender system to determine how collected data can best be utilised to reach pre-defined optimisation goals.[46]

46 To reach the optimisation goal, the algorithm can emphasise different ways of how to prioritise the collected data. For example, an algorithm could favour recency of a news article. Another strategy would be to look at popularity of articles as a ranking criterion of how to sort the final recommendations calculated for each user. Accuracy is another frequently used way of curating content. An accuracy-optimised approach tries to model user preference as closely as possible. They calculate recommendations that fall in line with existing user preferences. Depending on the data that was collected and the nature of the item in question, there are multiple ways of how to refine each strategy.

All large internet intermediaries, and social media platforms in particular, use so-called "open content recommender systems".[47] These systems utilise user-generated content in the recommender's source pool by default, but certain content items can be excluded based on, for instance, a violation of Terms of Service. To optimise engagement, these systems "personalise" the online experience by prioritising content that is assumed to be appealing and is generated through each individual's prior engagement and behaviour. Accordingly, the videos, search results, news articles or any other type of content that is displayed to the user is unique to their experience and differs from what other users see. For this reason, among others, algorithmic content recommender systems have the potential to undermine and disrupt democratic processes.[48] They risk narrowing individuals' exposure and access to different points of view, values and narratives, thereby threatening pluralism and diversity. This may necessitate state intervention and attention.

The algorithmic filtering and adaption of online content based on speculated personal preferences and interests decreases the exposure to a diversity of information, with potential negative effects on diversity and public discourse, as well as privacy. Content curation systems can therefore profoundly influence the information sources that form the basis for arriving at well-informed opinions, and thus the thought process of individuals. While the research is not yet conclusive, this could undermine individuals' ability to form their opinions and make them vulnerable to manipulative interference. As the systems are built on intrusive data practices and persuasion architectures (at scale), the inevitable personalisation of content might have a significant effect on the cognitive autonomy of individuals and interfere with their right to form an opinion.

47 As opposed to a closed recommender system that provides items to users from a limited list of options. These lists are curated by the platform owner.

48 N. Helberger (2019) On the Democratic Role of News Recommenders. Digital Journalism 7(8). Routledge: 993–1012. DOI: 10.1080/21670811.2019.1623700.

2.2 Curation and prioritisation of public interest content

The methods by which online platforms curate content through recommender systems are not transparent, and they are very rarely subject to public and/or state scrutiny. When internet intermediaries incorporate diversity into recommender systems, it is typically a design choice to engage users and increase profits. Platform curated diversity has primarily been utilised to optimise financial gain, rather than to promote democratic debate, through a practice of prolonged engagement to achieve what is referred to as an optimisation goal—increased ad revenue or a higher platform/service valuation through increased traffic.[49] Put simply, business-driven content curation benchmarks have largely been established to optimise economic gain and leverage user engagement for corporate interest, rather than seeking to reflect and ensure genuinely diverse content.[50]

Content recommender systems may also have unintended consequences from the perspective of broader societal objectives and can negatively shape and interfere with the absolute right to freedom of thought and opinion.

In addition, the processes of internet intermediaries' recommender systems typically exclude individual users' choice, control and agency—prerequisites to ensuring individual autonomy in seeking and imparting a variety of information and ideas. Following the public disclosure of a number of vulnerabilities of recommender systems,[51] there has been increasing public and state pressure to ensure that their processes better and more meaningfully prioritise "diversified" media exposure. In particular, concerns have been raised in the context of the legality and reach of political speech, and the spread and normalisation of specific value systems as protected speech, even when content is in violation of Terms of Services or international human rights standards. Given the complete lack of information on the way in which platforms govern and prioritise

49 In the words of a Facebook official: "Facebook is profitable only because when you add up a lot of tiny interactions worth nothing, it is suddenly worth billions of dollars.", K. Klonick, New Governor, p.1627.
50 K. Klonick, The New Governors, p.1664.
51 A well-known and most cited case was that of the "Napalm Girl", a journalistic photo which was taken down by Facebook based on its nudity policies.

speech, it is clear that recommender systems and correlated logics of optimisation could undermine the "fair opportunity to participate" [52] for all. At the same time, it should be recognised that human rights-based recommender systems can positively affect pluralism, for example in the contexts of authoritarianism and media capture.

It is important to define what constitutes public interest with regards to algorithmic content curation, and how it affects prioritisation of different types of content. The concept and definition of public interest content is as highly contested as the definition of diversified exposure. In principle, public interest content constitutes that information that "the public would have an interest in being informed about".[53] Another way to think about public interest content is as content that is relevant to the well-being of citizens, the life of the community or the local population. Obvious examples include COVID-19 pandemic information or information related to democratic voting processes. The volume of related, and not always reliable, content has prompted intermediaries to (publicly) prioritise accuracy. Several platforms have, in a comparatively short amount of time, demonstrated their capacity to re-configure algorithm recommender systems in an effort to filter out, or label false information, and prioritise content from trusted public health authorities. The success of these efforts, or logic behind the motivation for these changes, however, remains hotly debated,[54] while a lack of transparency into the underlying data and content moderation choices made by the platforms remain a mystery. Debates aside, internet intermediaries and social media platforms—facing increasing demands from the public and states that they be held responsible for public health awareness—have shown their capacity for reflection, and for restructuring how they prioritise and rank content.[55]

52 K. Klonick, The New Governors, p.1664.
53 M.E. Mazzoli and D. Tambini, Prioritisation uncovered: The Discoverability of Public Interest Content Online. Council of Europe (2020), p. 13.
54 M. Cinelli, The COVID-19 social media infodemic, The Nature (2020), p.10; See also: Global Disinformation Index, Why is tech not defunding COVID-19 disinfo sites? (2020), Retrieved from: https://disinformationindex.org/2020/05/why-is-tech-not-defunding-covid-19-disinfo-sites.
55 European Commission, Joint communication to the European Parliament, the European Council, The Council, The European Economic and Social Committee and the Committee of the Regions, Tackling COVID-19 disinformation - Getting the facts right, JOIN(2020) 8 final, 10 June 2020., section 5.

This points to a need for greater public attention and political pressure on platforms to make transparent their recommender processes and restructure recommender systems and their optimisation goals, in order to address structural problems of our contemporary media environment. The issue goes beyond questions of content governance, and additionally concerns competition law, media ownership and concentration rules.[56] It also highlights the urgent need to prioritise media pluralism and diversity policy objectives and interventions for a more enabling digital space.

2.3 News aggregation and media plurality

News aggregators function as a central hub of online news distribution, directing readers to news items, and other content deemed (by the news aggregator) to be news. This process is predominantly carried out by algorithms, which is why news aggregators are sometimes referred to as "algorithmic gatekeepers".[57]

News aggregators often involve a tension between "algorithmic logic" and "editorial logic".[58] "Algorithmic logic" significantly impacts diversity as well as political discourse by prioritising novelty, for example, over other criteria of newsworthiness (e.g. public relevance, diversity, etc.). A study of content curation processes behind AppleNews, which employs both human moderation (in the Top Stories) and algorithmic content curation (in Trending Stories), showed that human moderated content featured "more diverse and more equitable source distribution than algorithmically-selected" stories.[59] Trending Stories, according to the same study, almost exclusively included "soft news" (e.g. stories about celebrities), with Top Stories reserved for "hard news" (e.g. political content).[60] These practices were found to severely impact source plurality, and news distribution, and therefore content plurality, in two ways: First,

56 M. E. Mazzoli and D. Tambini, Prioritisation uncovered: The Discoverability of Public Interest Content Online. Council of Europe (2020), p. 23.
57 Napoli 2014.
58 T. Gillespie, PJ Boczkowski, KA Foot, Media technologies: Essays on communication, materiality, and society, MIT Press (2014).
59 J. Bandy and N. Diakopoulos, Auditing News Curation Systems: A Case Study Examining Algorithmic and Editorial Logic in Apple News, Proceedings of the Fourteenth International AAAI Conference on Web and Social Media (ICWSM 2020), p.43.
60 Ibid.

aggregators created a so-called “market-expansion effect” because they provided individuals exposure to news outlets with lower popularity or brand awareness. Second, the deployment of aggregators incentivised some users to limit or stop direct use of news outlets, resulting in the so-called “substitution effect”. Since user reactions are usually based on first impressions, clickbait is used in the news feed to attract attention and engage users, thereby facilitating advertising that generates profit. This approach further challenges media sustainability, and consequently independence and pluralism, adding to the overall pressure and financial constraints faced by legacy media because of internet intermediaries’ concentrated advertising and data exploitation models.

There is a growing imbalance between the outreach and communications impact of legacy media and online platforms, and content creation vs. content curation. Given that the traditional subscription-based business model is in decline, legacy media are struggling for viability. An ever-growing number of people get their news exclusively from other sources, where articles are more likely to be available “for free”. The willingness to pay for quality news has decreased, while usage of “free” news aggregator sites and social media platforms increased. As a result, many online news outlets have no choice but to search out new revenue streams. They are coerced into adopting many of the practices used by the large platforms, which exert significant power over the logic of the digital adtech industry (e.g., by employing targeted advertising, publishing sponsored content, or collecting and selling user data). This trend has profoundly negative implications for media freedom globally: It creates an environment in which legacy media organisations must compete with social media companies and intermediaries for the same revenue sources, while also being subject to intermediaries’ recommender systems and content curation policies. The situation contributes to an erosion of trust in media, decreased responsibility for creations and dissemination of disinformation and other problematic content.

Some legacy media organisations also employ algorithm-driven tools themselves, with content personalisation and optimisation playing integral roles in media production processes. There remain stark differences between "news logic of personalisation"[61] and "platform logic of personalisation", with news media subject to a systemic lack of technological and financial resources, devaluation of traditional editorial and professional ethics, the prevalence of newly emerging private interests, and economic incentives. Algorithmic content curation models and recommender strategies developed and deployed by independent legacy media outlets, and especially public service broadcasters, could offer alternative models for better ensuring audiences' exposure to diversity, potentially even offering individuals with models for "diversity by design".[62]

Evidence shows that journalists value "editorial logic", such as "transparency, diversity, editorial autonomy, broad information offer, personal relevance, usability, and surprise" over the business-driven algorithmic logic of recommender systems.[63]

Algorithm-driven content curation and recommendation processes and practices pose threats to media plurality and diversity and raise concerns regarding the full enjoyment of the right to freedom of expression. The major contributing factors constituting these threats are:

- Financial instability of, and fiscal pressure on, legacy media: Online platforms have gained enormous economic power, primarily through advertising revenue, and they use this leverage to dictate conditions

61 B. Bodó, Selling News to Audiences – A Qualitative Inquiry into the Emerging Logics of Algorithmic News Personalization in European Quality News Media, Digital Journalism (2019), p.17-18.
62 See more about this content in N.Helberger, Diversity by design—Diversity of content in the digital age, Government of Canada (2020), p.8; Natali Helberger, Kari Karppinen & Lucia D'Acunto (2018) Exposure diversity as a design principle for recommender systems, Information, Communication & Society, 21:2, 191-207, DOI: 10.1080/1369118X.2016.1271900.
63 This study involved newsrooms from the Netherlands and Switzerland; M. Bastian, N.Helberger & M. Makhortykh, Safeguarding the Journalistic DNA: Attitudes towards the Role of Professional Values in Algorithmic News Recommender Designs, Digital Journalism (2021), p.21.

for the curation of all online content, including editorial media and news content. This power imbalance includes an imbalance of "opinion power"[64] and the power to "influence processes of individual and public opinion formation", which in turn enables "these platforms [to] change the very structure and balance of the media market, and thereby directly and permanently impact the pluralistic public sphere."[65]

- Legacy media, compelled to adopt similar business models and social media "logic", miss an opportunity to change the "rules of the new communication orders",[66] and to contribute to a more diversified media landscape. Yet there are alternative algorithmic-curation models centring on public interest content and professional journalism practices—typically instated by legacy and public service media organisations—and they do offer alternative models to mitigate the potential problems caused by a lack of prioritisation of public interest content.[67]

- Improving algorithmic content curation with a goal of increasing diversity of media sources poses several challenges:

    - Even if internet intermediaries and social media companies "train and game" algorithms "for good"—to expose heterogeneous audiences to heterogeneous content—these practices lack meaningful transparency, and individuals have no agency regarding the design and logic that govern these systems. This presents a significant and systemic risk to the enjoyment of freedom of expression.

    - While personalised content and optimisation processes could contribute to meeting diverse individual, group and societal needs—and generate potential for diversity—such potential should be driven by public policy objectives and corresponding interventions.

64 N. Helberger, The Political Power of Platforms: How Current Attempts to Regulate Misinformation Amplify Opinion Power. Digital Journalism, 8(6), 842-854 (2020)
65 Ibid., p.846.
66 For an in-depth discussion about this problem, see: N. Helberger,The Political Power of Platforms: How Current Attempts to Regulate Misinformation Amplify Opinion Power. Digital Journalism, 8(6), 842-854 (2020)
67 M.E,Mazzoli and D.Tambini,Prioritisation uncovered: The Discoverability of Public Interest Content Online. Council of Europe (2020).

- There is little to no information on how content produced by and for marginalised communities circulates online, and how recommenders treat such content. Studies suggest[68] that certain content and speech is treated differently, giving rise to concerns that content is not equally accessible, and that safeguards to prevent discriminatory algorithmic outcomes and ensure fair and equal public participation and deliberation have not been developed or implemented.

- Consequences of algorithm-driven content curation can compound human rights abuses and rule of law violations when amplified in certain national contexts, specifically in conjunction with systemic (state-led and private) media capture and monopolised control of public dialogue. Under these circumstances, additional layers of algorithm-driven restrictions to and for media pluralism and diversity intensify the aggregate sum of individual loss of the right to freedom of expression.

68 See, for example: A. Chinmayi, Facebook's Faces, Forthcoming Harvard Law Review Forum Volume 135 (2021) and K. Klonick, The New Governors: The People, Rules, and Processes Governing Online Speech, The Harvard Law Review (2018); C. O'Neil, Facebook's VIP "Whitelist" Reveals Two Big Problems, Bloomberg Opinion (2021), Retrieved from:https://www.bloomberg.com/opinion/articles/2021-09-15/facebook-s-xcheck-vip-whitelist-reveals-two-big-problems